

CAD-to-Cost — Before / After the Audit Fixes

Basis: 200,000/yr · China · 5-year programme · every part driven through the REAL browser UI (Playwright), decisions answered by the engineer, blocking sanity acknowledged where demanded. Independent OCCT ground truth throughout. Generated by scripts/cad-audit-compare.py — the machine wrote this table from the captures; nothing here is hand-typed.

Per-part comparison

Part (truth)	BEFORE — costed as shipped	AFTER — costed on the fixed engine	Decisions answered (the fix working)	Audit findings it proves
Casting Bracket 320.0 cm³ · cast iron → expect 2.288 kg	¥204.66 EN-GJS-500-7 (Ductile Iron) net 2.359 kg · measured 2.272 kg	¥189.13 EN-GJL-250 net 2.359 kg · measured 2.304 kg	material.family=cast iron	F1/F8 + gap 1: was aluminium LM25 @ its own mass; grade now cast iron; F9 tooling
Steering Knuckle 356.1 cm³ · cast iron → expect 2.546 kg	no costed run (crashed / timed out / gated)	¥198.06 EN-GJL-250 net 2.625 kg · measured 2.564 kg	material.family=cast iron	F7 — the browser died silently before; now costs
Stub Axle PRCR002 1037.1 cm³ · steel → expect 8.141 kg	¥399.52 1020 / S20C net 8.141 kg · measured 8.141 kg	¥391.31 1020 / S20C net 8.141 kg · measured 8.141 kg	material.family=steel	F3 — auto run said sheet_metal before
Part1 (casting) 429.4 cm³ · cast iron → expect 3.070 kg	¥281.52 LM25 / A356 net 1.195 kg · measured 1.151 kg	¥301.97 EN-GJL-250 net 3.165 kg · measured 3.092 kg	material.family=cast iron	gap 1 — costed default LM25 @ aluminium mass, silent, before
Seat Locking Bracket 71.1 cm³ · steel → expect 0.558 kg	¥15.87 DC01 net 0.558 kg · measured 0.558 kg	¥16.25 DC01 net 0.558 kg · measured 0.558 kg	material.family=steel	F4 — blank was 210% of the part before
Flange (fixture) 63.9 cm³ · steel → expect 0.502 kg	¥177.31 6061-T6 net 0.502 kg · measured 0.173 kg	¥149.77 1045 / C45 / 080M46 (Medium-Carbon Steel) net 0.502 kg · measured 0.502 kg	material.family=steel	F1 — was mat-al6061 @ steel mass before
Block (fixture) 44.9 cm³ · steel → expect 0.352 kg	no costed run (crashed / timed out / gated)	¥74.17 1045 / C45 / 080M46 (Medium-Carbon Steel) net 0.352 kg · measured 0.352 kg	material.family=steel	G5 family token resolution
Bumper 2059.9 cm³ · pp → expect 1.854 kg	no costed run (crashed / timed out / gated)	API capture (no headless total) mat-pp-impact net 1.8539 kg · measured 1.8539 kg	—	F2 — forced run took aluminium mass before (API capture; UI times out)
Fuel Tank 10595.9 cm³ · hdpe → expect 10.172 kg	no costed run (crashed / timed out / gated)	API capture (no headless total) mat-hdpe-fuel-coex net 10.1721 kg · measured 10.1721 kg	—	sealed-void plastic (API capture; UI times out)

Closing machine-checks (23/23 pass)

Check	Verdict	Evidence
C1 Casting Bracket	PASS	answered: material.family=cast iron
C3 Casting Bracket	PASS	report net 2.359 kg vs family-density mass 2.304 kg (Δ2.4%) — within the documented cast-iron density-rounding residual
C4 Casting Bracket	PASS	net 2.359 kg vs 320.0 cm³ × 7.15 (cast iron) = 2.288 kg (Δ3.1%)
C1 Steering Knuckle	PASS	answered: material.family=cast iron

C3 Steering Knuckle	PASS	report net 2.625 kg vs family-density mass 2.564 kg ($\Delta 2.4\%$) — within the documented cast-iron density-rounding residual
C4 Steering Knuckle	PASS	net 2.625 kg vs $356.1 \text{ cm}^3 \times 7.15$ (cast iron) = 2.546 kg ($\Delta 3.1\%$)
C1 Stub Axle PRCR002	PASS	answered: material.family=steel
C3 Stub Axle PRCR002	PASS	report net 8.141 kg vs family-density mass 8.141 kg ($\Delta 0.0\%$)
C4 Stub Axle PRCR002	PASS	net 8.141 kg vs $1037.1 \text{ cm}^3 \times 7.85$ (steel) = 8.141 kg ($\Delta 0.0\%$)
C1 Part1 (casting)	PASS	answered: material.family=cast iron
C3 Part1 (casting)	PASS	report net 3.165 kg vs family-density mass 3.092 kg ($\Delta 2.4\%$) — within the documented cast-iron density-rounding residual
C4 Part1 (casting)	PASS	net 3.165 kg vs $429.4 \text{ cm}^3 \times 7.15$ (cast iron) = 3.070 kg ($\Delta 3.1\%$)
C1 Seat Locking Bracket	PASS	answered: material.family=steel
C3 Seat Locking Bracket	PASS	report net 0.558 kg vs family-density mass 0.558 kg ($\Delta 0.0\%$)
C4 Seat Locking Bracket	PASS	net 0.558 kg vs $71.1 \text{ cm}^3 \times 7.85$ (steel) = 0.558 kg ($\Delta 0.0\%$)
C1 Flange (fixture)	PASS	answered: material.family=steel
C3 Flange (fixture)	PASS	report net 0.502 kg vs family-density mass 0.502 kg ($\Delta 0.0\%$)
C4 Flange (fixture)	PASS	net 0.502 kg vs $63.9 \text{ cm}^3 \times 7.85$ (steel) = 0.502 kg ($\Delta 0.0\%$)
C1 Block (fixture)	PASS	answered: material.family=steel
C3 Block (fixture)	PASS	report net 0.352 kg vs family-density mass 0.352 kg ($\Delta 0.0\%$)
C4 Block (fixture)	PASS	net 0.352 kg vs $44.9 \text{ cm}^3 \times 7.85$ (steel) = 0.352 kg ($\Delta 0.0\%$)
C4 Bumper	PASS	net 1.8539 kg vs $2059.9 \text{ cm}^3 \times 0.9$ (pp) = 1.854 kg ($\Delta 0.0\%$)
C4 Fuel Tank	PASS	net 10.1721 kg vs $10595.9 \text{ cm}^3 \times 0.96$ (hdpe) = 10.172 kg ($\Delta 0.0\%$)

Reading the table

A green AFTER cell means: alloy and weight agree with the independent measurement (family density $\times$ measured volume), the PDF's two mass statements agree, and the material was explicitly confirmed rather than guessed. "No costed run" in BEFORE marks the parts the shipped product could not cost at all (browser crash F7, UI timeout) — the absence is the finding.